

Activation functions

for output layer (Rule)
Linear (units = 1)

for output layer (Rule)
if binary Classification:
① Sigmoid (units: 1)
② Softmax (units: 2)

if multi-class Classification:
Softmax (units: no of unique
class labels in
label column)

for hidden layer (Guideline)
✓ ① relu (90% problems solved)
✓ ② leakyrelu

avoid in deep NN { ③ sigmoid } VGP (Vanishing Gradient Problem)
 ④ tanh
gradients are nearing to ZERO
 $\Delta w \sim 0$ } no improvement
 $\Delta b \sim 0$ } learning is paused

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Optimizers: Its a backpropagation algo implementation used to calc Δw & Δb & update w & b after every epoch.

Guideline:

- Ⓐ Adam
- Ⓑ Nadam
- Ⓒ Adadelta
- Ⓓ Adamax
- Ⓔ sgd
- Ⓕ lion

Error functions

- * ① MSE (mean squared error)
- ② MAE (mean absolute error)

Metric functions

- ① r^2 score

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Error functions

if binary Classification:

binary_crossentropy

if multiClass Classification:

if labelCol is discreteND:

sparse_categorical_crossentropy

if labelCol is dummyVariables:

categorical_crossentropy

Metric function

if balancedDataset:

accuracy $\rightarrow$ ① generalization check
② CL check with entire dataset

if unbalancedDataset:

accuracy $\rightarrow$ generalization check.

f1 score
-or-
PR pair $\rightarrow$ CL check with entire dataset.

Guidelines for architecting an ANN

② for classification use-case:

Option 1:

No of hidden layers : no. of feature cols

No of units per hidden layer: no of feature cols

Option 2:

No of hidden layers : no. of feature cols

No of units per hidden layer: $3 * \text{no of feature cols}$

Option 3:

No of hidden layers : no. of feature cols

No of units per hidden layer: $\frac{1}{3} * \text{no of feature cols}$

for regression usecases:

No of hidden layers:

Option 1: no. of feature cols

Option 2: $3 * \text{no. of feature cols}$

Option 3: $\frac{1}{3} * \text{no of feature cols}$

No of units per hidden layer:

Always start with 100 units, and increment in multiples of 5.

Deal with Overfitting in ANN

Note: Overfitting might be good for some problem statements. Check with your domain expert if overfitted models are acceptable.

① Try changing the OPTIMIZER.

- Ⓐ Adam
 - Ⓑ Nadam
 - Ⓒ RMSProp
 - Ⓓ Adadelta
 - Ⓔ Lion
- } → Try this in default setting.

② Tweak the learning rate parameter in optimizer.

- Ⓐ Adam
 - Ⓑ Nadam
 - Ⓒ RMSProp
 - Ⓓ Adadelta
 - Ⓔ Lion
- } → learning_rate → 0 to 1 (3 point precision)
0.001
⋮
1

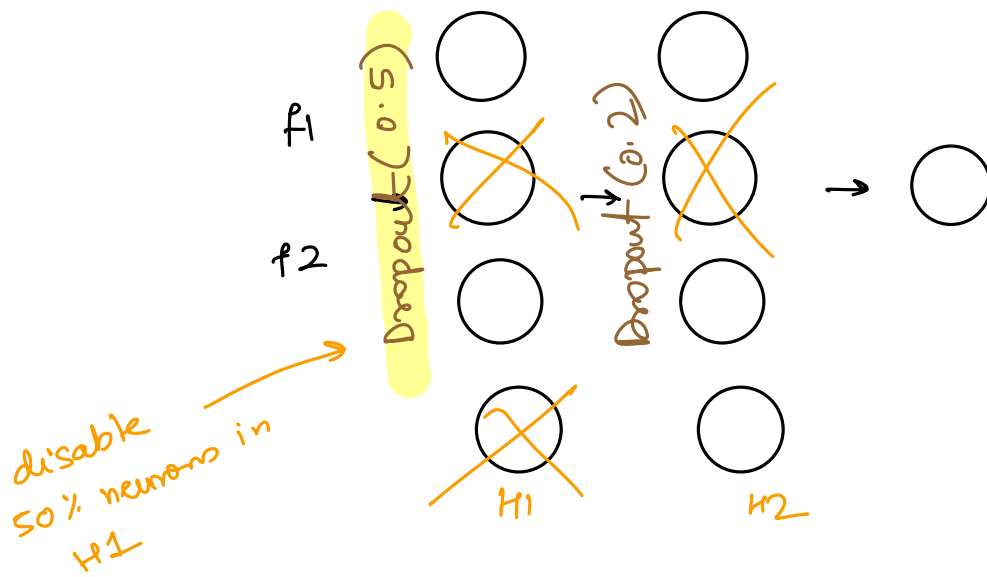
③ Tweak the momentum parameter in optimizer

- Ⓐ sgd
 - Ⓑ RMSprop
- } momentum → 0 to 1 (3 point precision)
0.001
⋮
1

④ Try regularization technique.

↳ adding penalty while calc Δw & Δb

② Dropout Regularization (randomly disabling the neurons in the dense layer)



③ L1 regularization

④ L2 regularization

⑤ Increase the data
— or —
change the ANN architecture

} Last option !